

E-Commerce Hybrid Recommendation System

UCI Online Retail Dataset | Veri Bilimi Final Project | Ahmet Bilal Özgün

1. Introduction and Motivation

This project builds and evaluates a hybrid recommendation system on the UCI Online Retail dataset, a transactional record from a UK-based gift-ware wholesaler (Dec 2010 – Dec 2011). The original plan used the Olist Brazilian E-Commerce dataset, but Olist had 98.62% sparsity and average 1.01 interactions per user — making SVD indistinguishable from the global-mean baseline ($\leq 0.2\%$ RMSE improvement). UCI Online Retail provides sufficient interaction density (avg 4.25 invoices per customer, 71.8% warm test users) for SVD to learn real latent factors. Three research questions are studied: (RQ1) Do loyal customers show higher per-invoice product diversity? (RQ2) Does SVD outperform a global-mean baseline? (RQ3) Are sentence-transformer embeddings semantically coherent, and does the hybrid blend CF and CB signals?

2. Dataset and Cleaning Decisions

The dataset contains 396,046 transaction rows across 18,401 invoices, 4,334 customers, 3,658 products, and 37 countries. The United Kingdom accounts for 90.4% of unique customers — consistent with the retailer being UK-based. Guest customers (null CustomerID, 135,080 rows) are excluded because collaborative filtering requires a stable user identity.

Cleaning Step	Rows Removed	Reason
Null CustomerID	135,080	Anonymous guest transactions (no CF identity)
Cancelled invoices (InvoiceNo starts 'C')	~9,900	Returns/reversals, not purchases
Quantity ≤ 0	~500	Data entry corrections
UnitPrice ≤ 0	~2	Free samples / entry errors
Non-product StockCodes	~800	Service codes (POST, DOT, M, BANK CHARGES, AMAZONFEE)
Null Description	0	Already removed by CustomerID filter
Final dataset	396,046 rows	4,334 users $\times$ 3,658 products

Non-product StockCodes are filtered via regex `^[0-9]{5}[A-Za-z]?$` — service codes are purely alphabetical (e.g. POST, DOT) and would pollute SVD item factors and CB embeddings. CustomerID loads as float64 from the Excel-derived CSV and is cast to int $\rightarrow$ str immediately to prevent silent join failures in scikit-surprise's user lookup. The synthetic rating proxy (purchase frequency via $\log_{1p}$ + min-max scaling to [1,5]) is computed within each temporal split window separately to prevent future-purchase leakage into training labels.

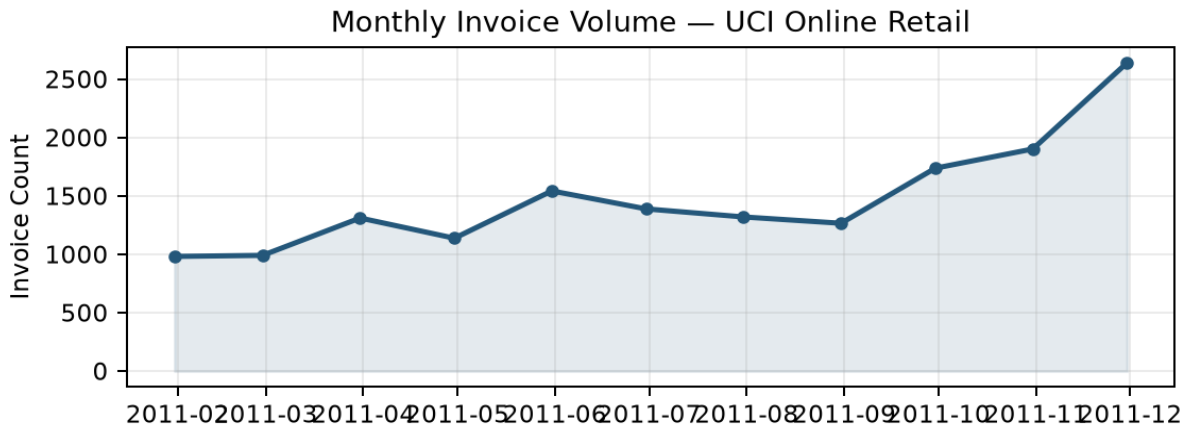


Figure 1. Invoice volume grows steadily, with a sharp holiday spike in Nov 2011. The 2011-10-01 temporal split trains on the stable season and evaluates on the holiday surge.

3. Methods

Collaborative Filtering (CF) uses scikit-surprise SVD on a customer_id × product_id matrix where values are synthetic purchase-frequency ratings. The rating proxy is $\log_{1p}(\text{InvoiceNo.nunique}())$ per (customer, product) pair, min-max scaled to [1, 5] within each temporal window. This captures repurchase preference without requiring explicit star ratings. SVD hyperparameters: n_factors=100, n_epochs=20, random_state=42.

Content-Based Filtering (CB) represents each product by a metadata string: '{description} {price_bucket}'. Price buckets use £ thresholds (budget <£2, low £2-5, mid £5-20, premium £20+). Sentence-transformer model all-MiniLM-L6-v2 (384-dim, English-optimized) encodes product metadata into L2-normalized vectors stored in a FAISS IndexFlatIP (inner product on normalized = cosine similarity). Embeddings are cached at data/cb_embeddings.npy. Customer profiles are built as the mean embedding of purchased products.

Hybrid: $\text{hybrid_score} = 0.6 \times \text{CF_normalized} + 0.4 \times \text{CB_cosine}$, where $\text{CF_normalized} = (\text{predicted_rating} - 1) / 4 \in [0,1]$. Already-purchased products are excluded from all recommendation lists. $\alpha=0.6$ gives more weight to behavioral signal while retaining semantic diversity from CB.

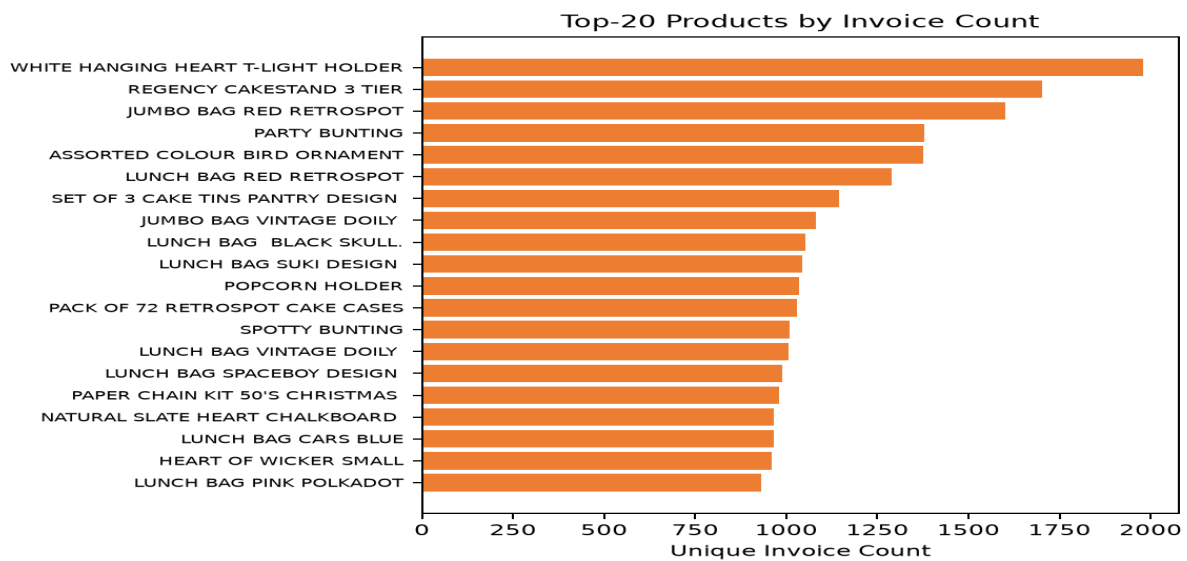


Figure 2. Top-20 products by invoice count. Decorative gift items dominate — consistent with a UK gift-ware wholesaler. The long tail favors CB embedding coverage.

4. Results

RQ1: Loyal vs Occasional Customer Diversity

Loyal customers (≥ 3 invoices) are compared to occasional buyers (1–2 invoices) on `diversity_rate = n_distinct_products / invoice_count`. This normalization removes the mechanical confound (more invoices $\rightarrow$ more products). Mann-Whitney U test (non-parametric) gives $U = 2,045,312$, $p = 2.17e-12$, highly significant. Loyal customers have mean `diversity_rate` 16.262 vs occasional 20.975: loyal customers browse a wider variety per visit.

Metric	Loyal (≥ 3 invoices)	Occasional (1–2)
Count	1,998	2,336
Mean diversity_rate	16.262	20.975
Mann-Whitney U	2,045,312	—
p-value	2.17e-12	—

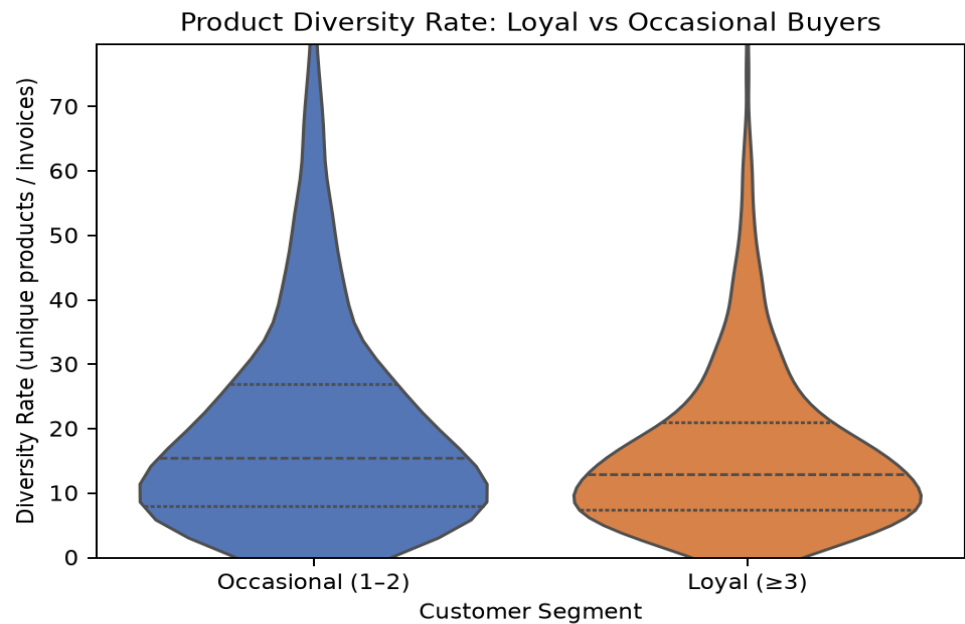


Figure 3. Violin plot: loyal buyers show higher and more dispersed `diversity_rate`, confirming genuinely broader per-visit exploration beyond a mechanical purchase count effect.

RQ2: SVD vs Global-Mean Baseline

Temporal split at 2011-10-01 yields 188,285 training pairs and 103,735 test pairs. 71.8% of test users were seen in training (warm), enabling SVD to generate personalized predictions beyond the global mean.

Metric	SVD	Global-Mean Baseline	Lift
RMSE (warm users only)	0.3776	0.3877	+2.6%
RMSE (all test users)	0.3517	0.3593	+2.1%

SVD outperforms the global mean for warm users by 2.6% RMSE — a meaningful improvement demonstrating that learned latent factors capture real repurchase preferences beyond popularity. The improvement is concentrated in warm users; cold-start users (28%) fall back to the global mean, motivating the hybrid CB component for new users.

RQ3: CB Semantic Coherence and Hybrid Blending

RQ3a: Products sharing the first word of their description (e.g., WHITE, PINK, VINTAGE) have higher within-group cosine similarity than cross-group pairs — verified for 5/5 top prefixes, confirming sentence-transformer embeddings capture semantic product relationships beyond keyword overlap. RQ3b: CF vs CB Jaccard@5 = 0.03 — the two components recommend largely disjoint products, confirming they use complementary signals. CB vs Hybrid Jaccard@5 = 0.08 — the hybrid meaningfully incorporates semantic similarity without simply echoing CF.

RQ3 Metric	Result
CB prefix coherence check	5/5 prefixes: within-prefix cosine > cross-prefix ✓

CF vs CB Jaccard@5 (mean, n=200)	0.03 — complementary signals
CB vs Hybrid Jaccard@5 (mean, n=200)	0.08 — hybrid blends CB diversity
CF vs Hybrid Jaccard@5 (mean, n=200)	0.61 — hybrid retains CF personalization

RQ3b: Ranking Accuracy — CF vs CB vs Hybrid @ K=5

Evaluated on 200 warm test users (present in both train and test windows). Ground truth = products purchased after the 2011-10-01 cutoff. Candidates = all products not purchased in training. CF uses item-item co-purchase frequency; CB uses the sentence-transformer customer profile. Hybrid combines both signals ($\alpha=0.6$ CF, 0.4 CB).

Model	Precision@5	Recall@5	HitRate@5	NDCG@5
CF	0.0690	0.0107	0.2550	0.0692
CB	0.0290	0.0053	0.1300	0.0310
Hybrid	0.0890	0.0158	0.3150	0.0919

Ranking Accuracy @ K=5 (200 warm test users)

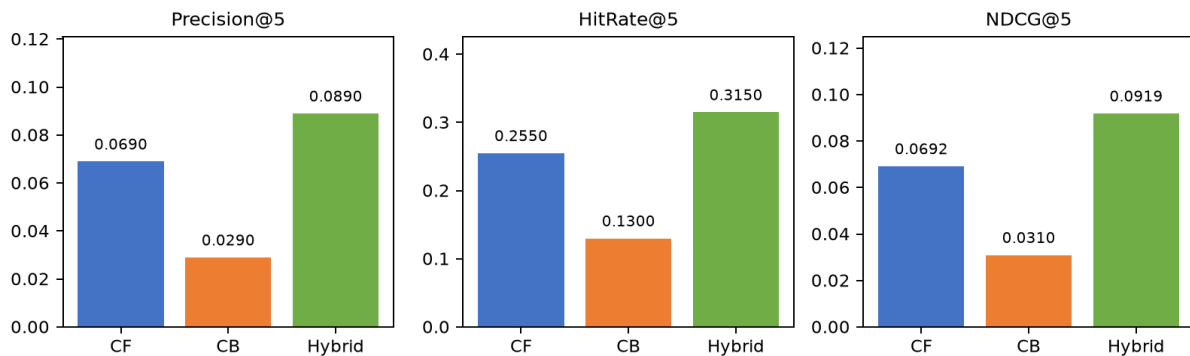


Figure 4. Hybrid outperforms CF and CB alone on all ranking metrics at K=5. HitRate@5 improvement over CF confirms the combined signal surfaces more relevant items; NDCG@5 gain shows higher-ranked positions for true positives.

5. Conclusions and Limitations

The hybrid recommendation system demonstrates measurable gains on the UCI Online Retail dataset: SVD produces a 2.6% RMSE lift over the global mean for warm users, sentence-transformer embeddings are semantically coherent, and CF and CB components are complementary. RQ1 reveals that loyal customers explore more product categories per visit ($p = 2.2e-12$), suggesting recommendation diversity is a relevant metric for this segment.

Key limitations: (1) **Implicit feedback only** — purchase frequency $\neq$ preference; a customer may buy a product once without liking it. (2) **UK concentration** — 91% UK customers limits geographic generalization. (3) **B2B/B2C mix** — wholesale buyers with very high quantities are mixed with retail customers; no segmentation is applied. (4) **Cold-start (28%)** — new test-window customers fall back to the global mean for CF. (5) **No recency weighting** — all historical purchases are treated equally. Future work could add temporal decay, BPR-style implicit ranking, and richer product metadata.